Youngjoon Jeong

Ph.D. Candidate @ Seoul National University
Physical AI / World Models

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Research Interest

Physical AI

- Representation learning for robot policy learning and world models, including latent-action and vision-language-action (VLA) models.
- Video/world models for visual planning, action-conditioned prediction, and efficient embodied decision-making.
- Robustness and generalization across viewpoints, tasks, environments, and robot embodiments.

Research Experience

Ph.D. Researcher @ Seoul National University — **Seoul, South Korea**
Advisor: Taesup Kim — Sep. 2023–Present
Selected projects:

- **Latent actions for robot policy learning**: VILA (CVPR 2026) and PoLAR (arXiv 2026): learn view-invariant and structured action representations from visual observations for downstream visuomotor policies.
- **Efficient visual world-model planning**: Sparse Imagination (ICLR 2026): reduce the computational cost of planning with imagined future visual states.
- **Robust vision-language-action policies**: J-PARC (arXiv 2026): study joint-level physical faults and adaptive action correction under altered robot dynamics.
- **Composable latent actions for visual world models (Ongoing)**: develop composable latent actions for controllable novel-view synthesis and video world models.

Education

Ph.D. Candidate @ Seoul National University — **Seoul, South Korea**
Advisor: Taesup Kim — Sep. 2023–Present

M.S. @ Seoul National University — **Seoul, South Korea**
Mar. 2020–Feb. 2022

B.S. @ Seoul National University — **Seoul, South Korea**
Mar. 2016–Feb. 2020

Publication

[1] **Youngjoon Jeong***, Junha Chun* & Taesup Kim. *Learning to Act Robustly with View-Invariant Latent Actions*. CVPR 2026. (* Equal contribution.) Project | Code

[2] Junha Chun*, **Youngjoon Jeong*** & Taesup Kim. *Sparse Imagination for Efficient Visual World Model Planning*. ICLR 2026. (* Equal contribution.) Project | Code

[3] **Youngjoon Jeong***, Junha Chun*, Soonwoo Cha & Taesup Kim. *Object-centric world model for language-guided manipulation*. ICLR World Model Workshop, 2025. (* Equal contribution.)

[4] **Youngjoon Jeong**, Jihwan Yu, Minsoo Jo, Junha Chun & Taesup Kim. *PoLAR: Factorizing Extent and Mode in Latent Actions for Robot Policy Learning*. arXiv preprint, 2026. Project | Code

[5] Minsoo Jo, Taeju Kwon, Junha Chun, **Youngjoon Jeong** & Taesup Kim. *Uncovering Vulnerability of Vision-Language-Action Models under Joint-Level Physical Faults*. arXiv preprint, 2026. arXiv